



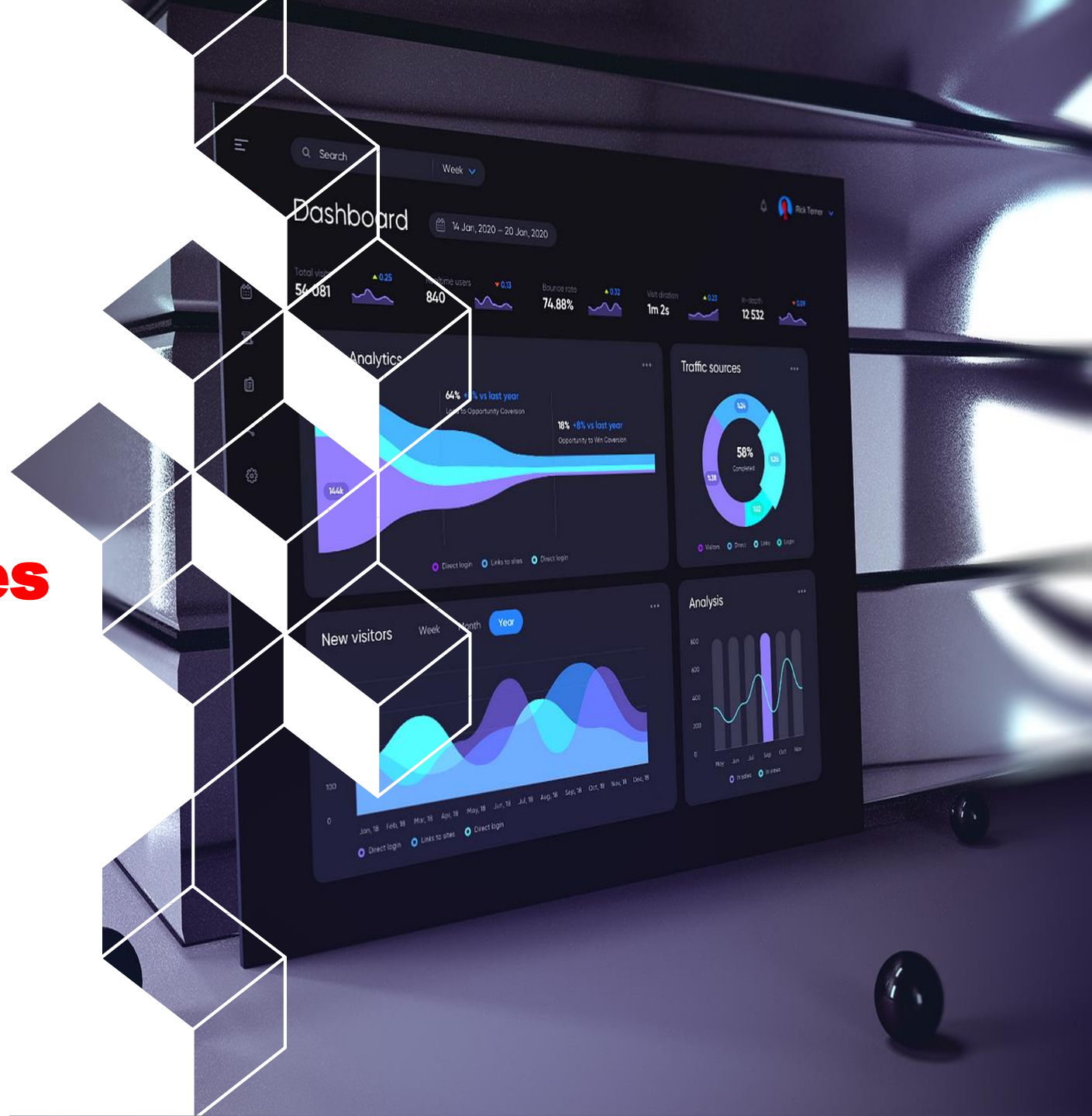
ICALEPCS 2023 Control GUI Strategies REX

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Goals of the workshop Controls GUI Strategies

Main link <https://indico.cern.ch/event/1299332/>

To gather, learn and discuss :

- Status & plans of GUI solutions used in CS across the world
- Charting solutions & data decimation
- Synoptics solution
- Web application packaging and deployment
- Zero/Low-code solutions

Page ICALEPCS <https://indico.cern.ch/event/1299332/contributions/>

Page [XWIKI](#)

GUI Situation & Strategy @ CERN

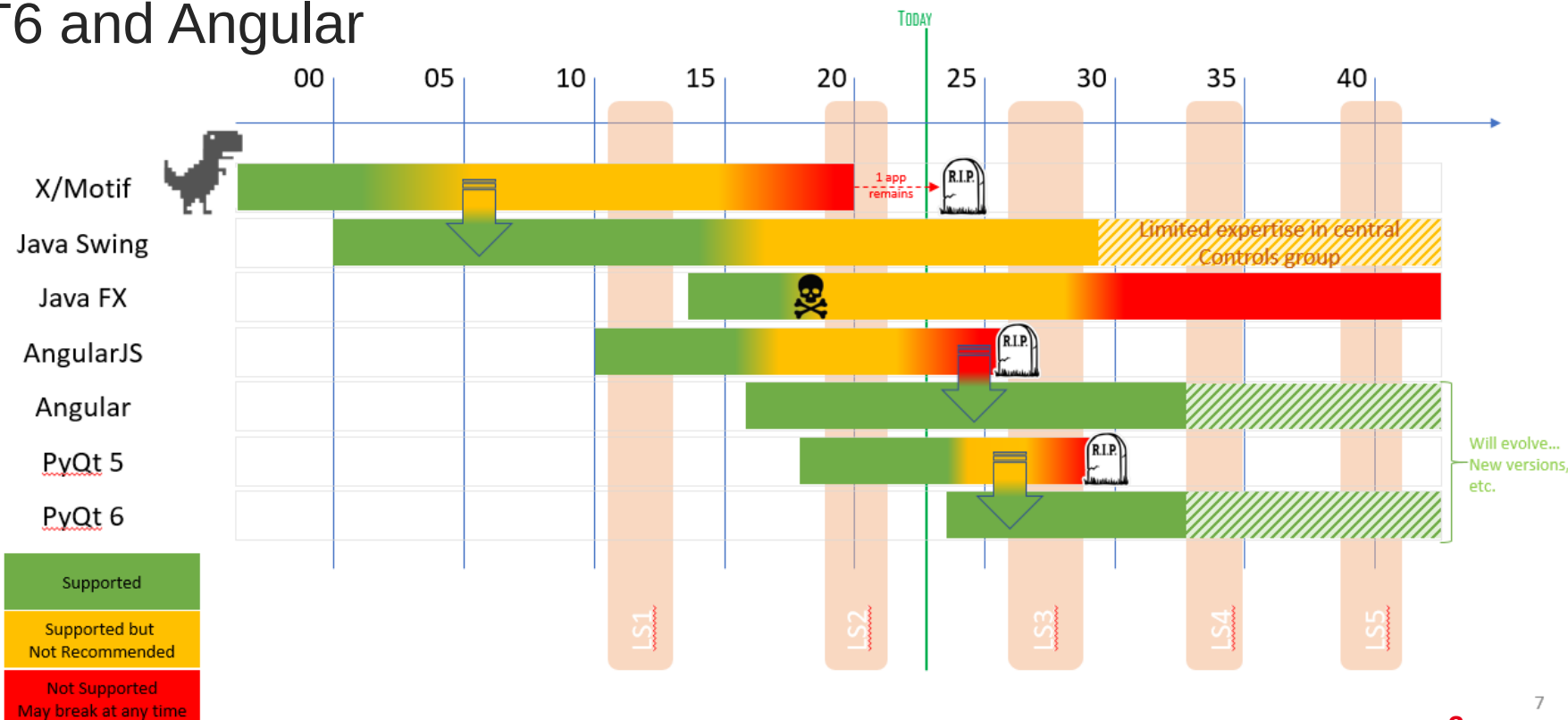
<https://indico.cern.ch/event/1299332/contributions/5617565/>

Speaker : stephane.deghaye@cern.ch



1. Avoid application breaks
 2. Maintain & extend applications
- Java Swing, PyQt6 and Angular

	Java Swing	Java FX
SY-ABT	✓	✓
SY-BI	✓	✓
BE-CEM	✓	✗
SY-EPC	✗	✗
TE-MPE	✓	✓
BE-OP	✓	✓
SY-RF	✓	✗

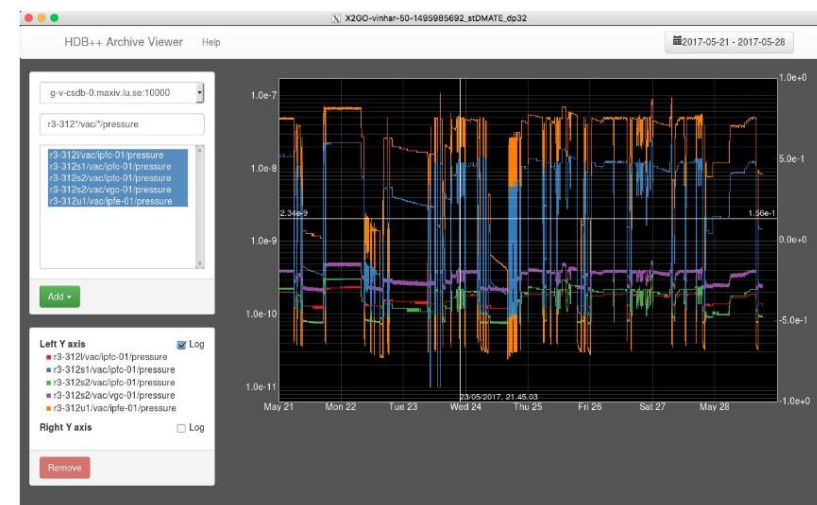
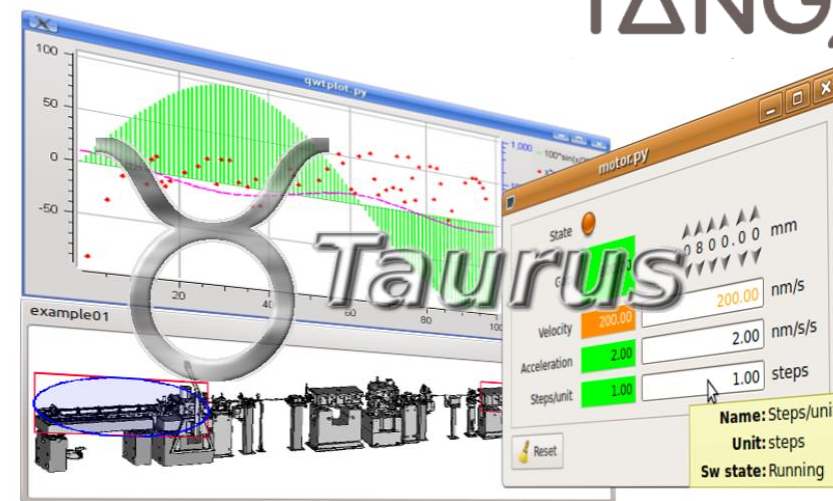
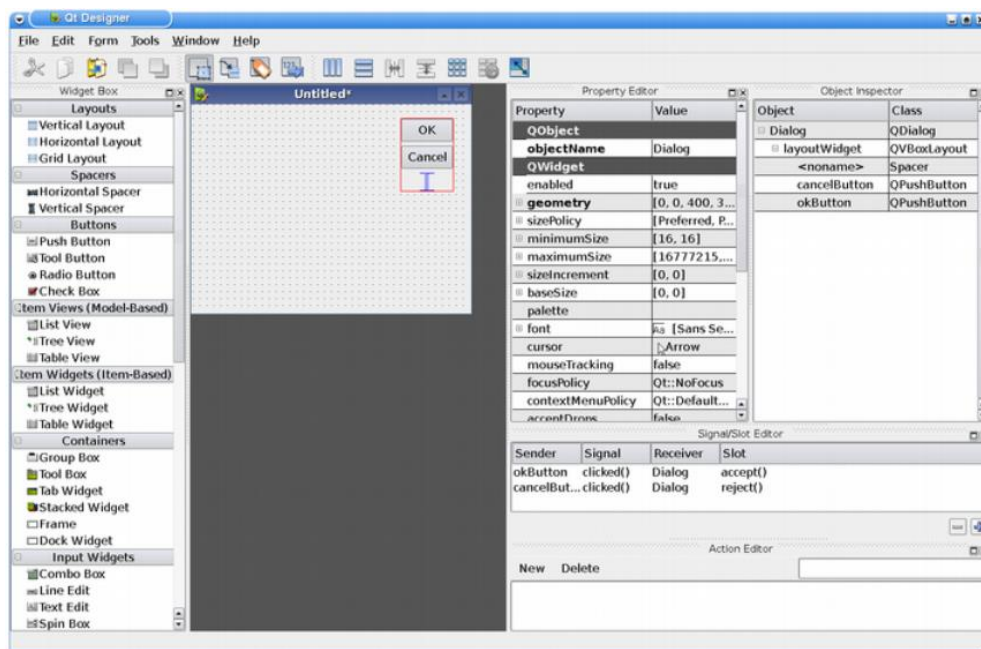


GUI Situation & Strategy @ MAX IV

<https://indico.cern.ch/event/1299332/contributions/5617566/>

Speaker : vincent.hardion@maxiv.lu.se

1. Increase the autonomy of their users
 - Low Code Application (Drag & Drop)
 - Taurus & Taranta Python QT Reac (Web)

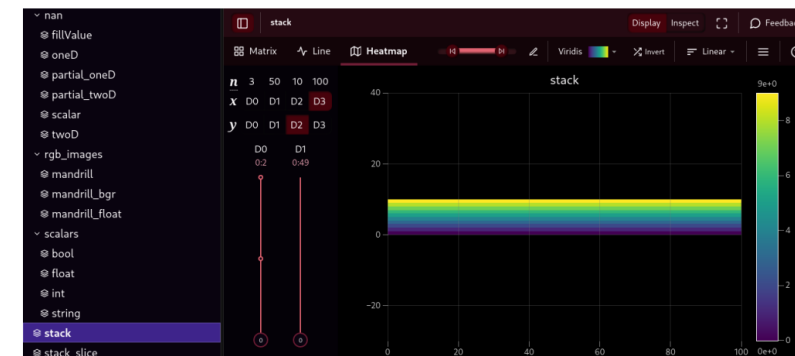
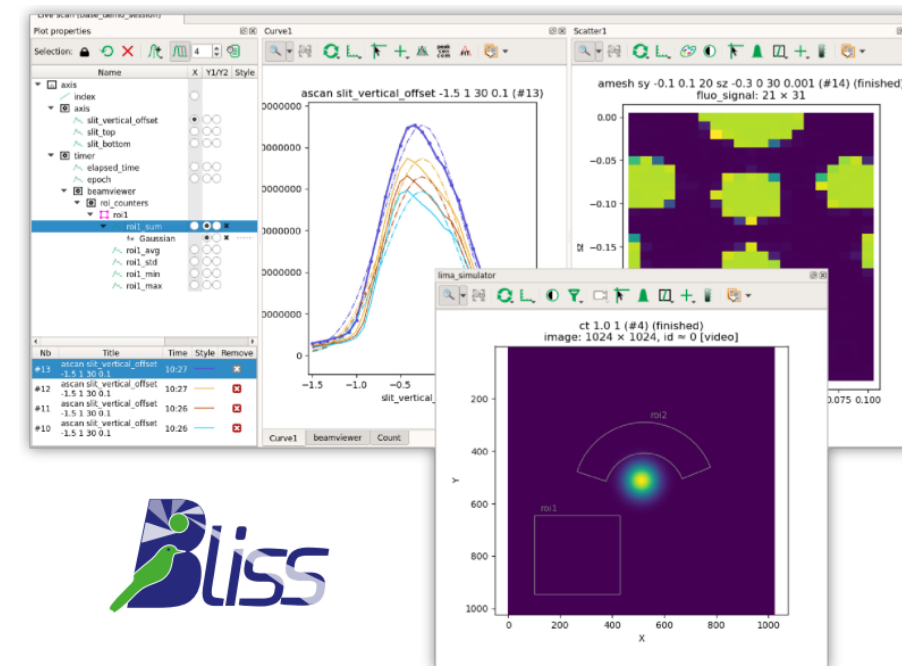
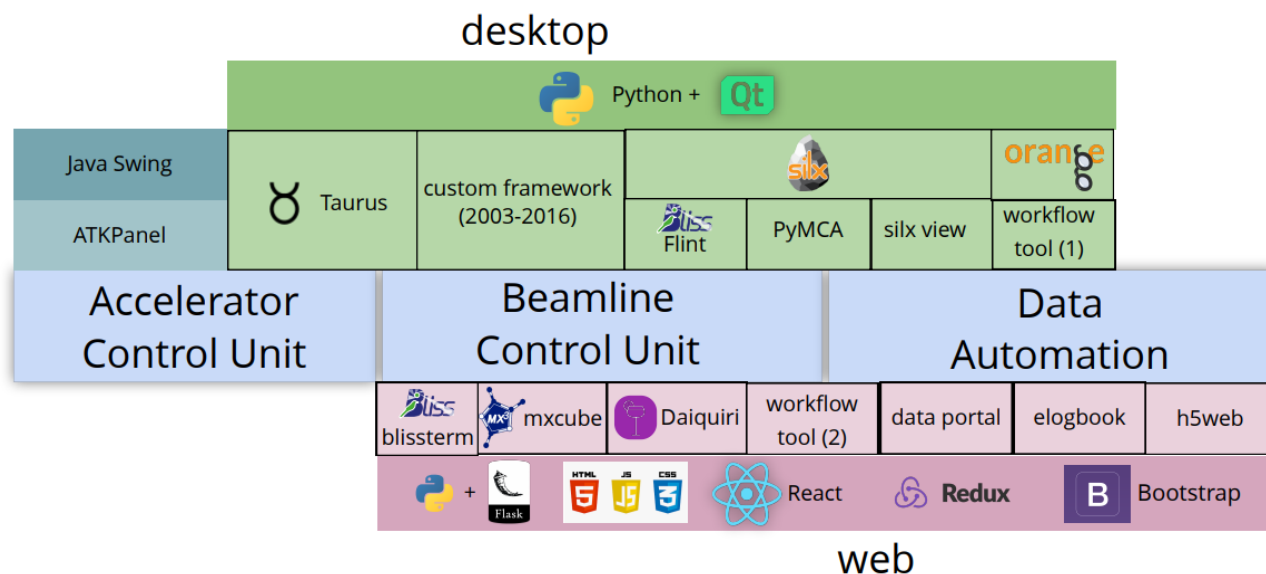


GUI Situation & Strategy @ ESRF

<https://indico.cern.ch/event/1299332/contributions/5618442/>

Speaker : matias.guijarro@esrf.fr

- No installation (Web solution)
- Accelerator Java Swing Rich client
- Beamlines (BLISS group) Python QT (Web)
- MXCube-Web <https://github.com/mxcube/mxcubeweb>
- Python flask, Rest API
- [H5web](#) Reac on HDF5 file format



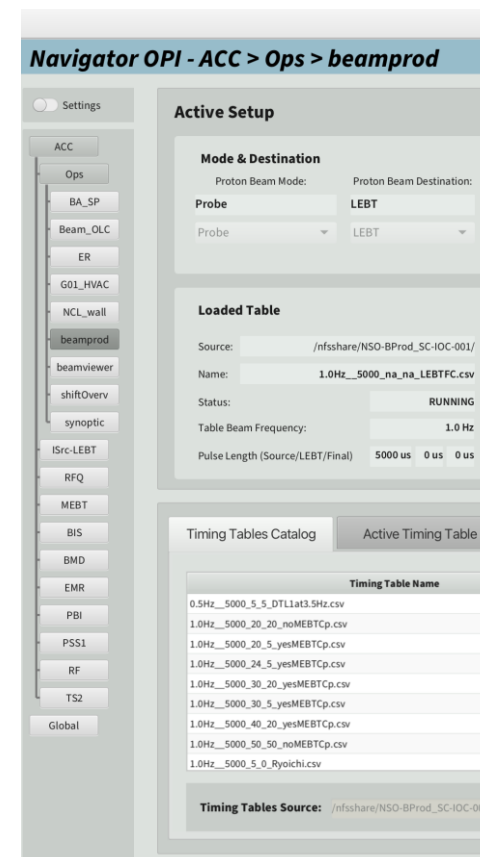
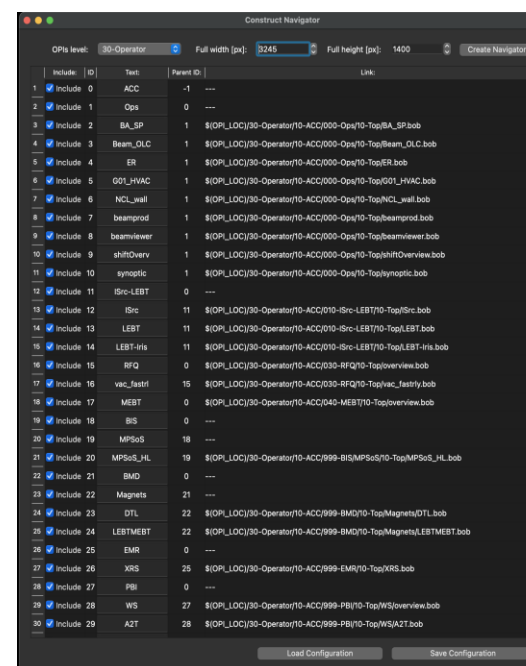
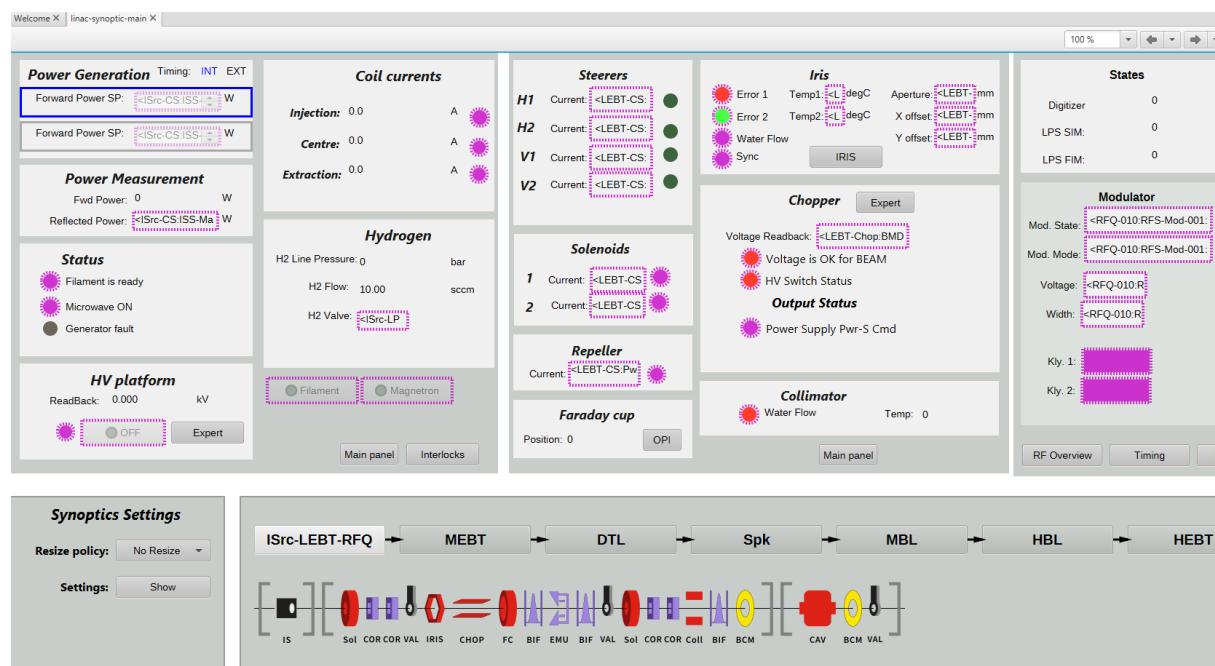
GUI Situation & Strategy @ ESS

<https://indico.cern.ch/event/1299332/contributions/5618443/>

Speaker : benjamin.bolling@ess.eu

Generate navigation menu & CS-Studio synoptic from python

- <https://gitlab.esss.lu.se/opis/navigator/tool>
- <https://gitlab.esss.lu.se/opis/acc/ops/operator>



GUI Situation & Strategy @ DIAMOND

Codathon March 2023

Mentor : Andy Wilson

Prototype a spec for the BOB file format

Web solution React + GraphQL

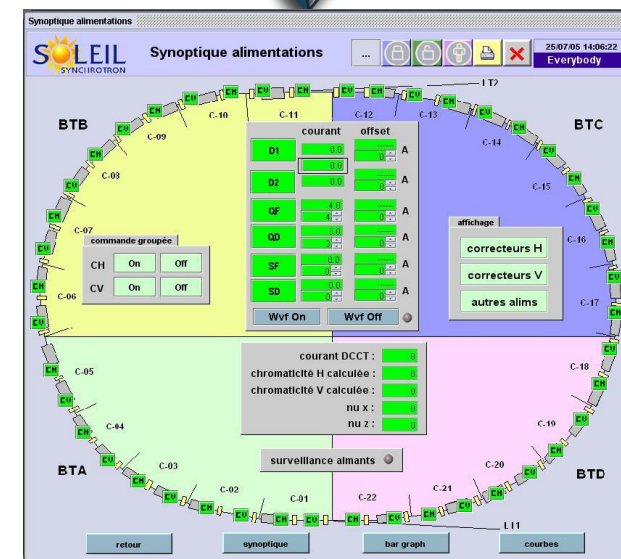
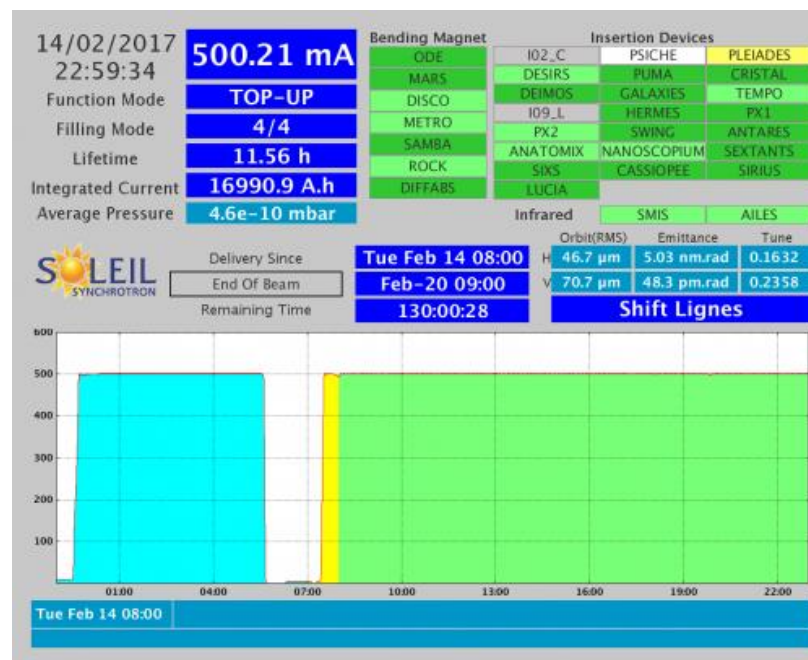
<https://github.com/dls-controls/cs-web-proto>





Users autonomy, central deployment

- 



GUI Situation & Strategy @ CEA IRFU

MUSCADE @ IRFU :



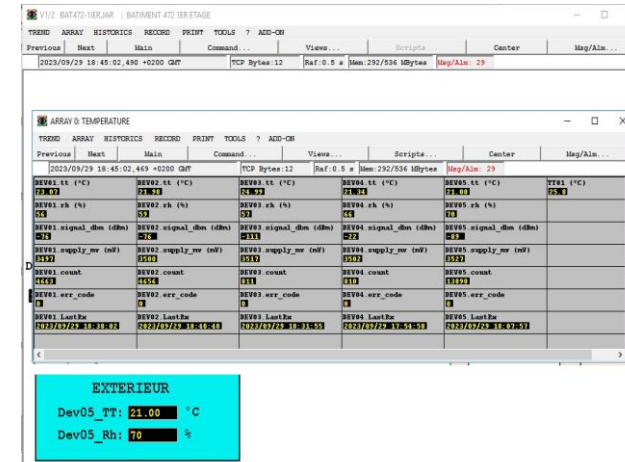
A AUTOCAD

- Java rich client application (ANIBUS)
- Android application for Smartphone
- Java Web Start

EPICS @ IRFU :

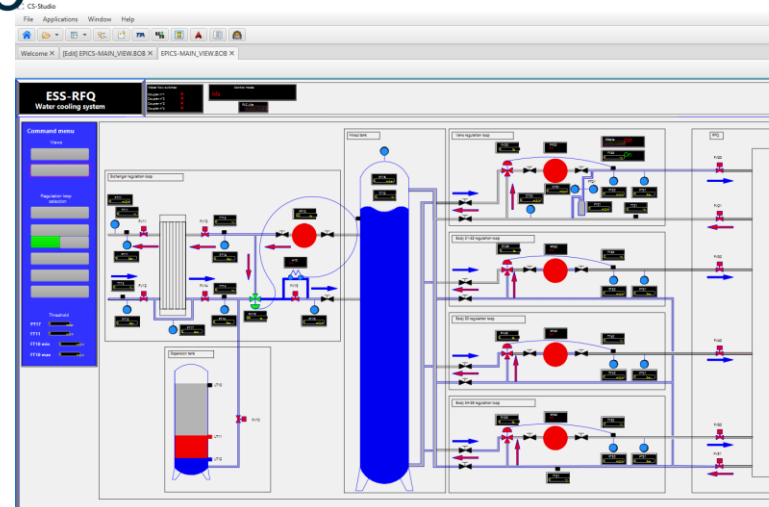


- Java rich client application CS-Studio
- DBWR – Web View (Tomcat & JSP)
- <http://132.166.10.46/phoebus/TT472/>



ISEULT Synthèses et alarme

AL_URGENT_MUSCADE	0	ACQUITTEMENT
Synthèse cryogénie		
Détails process	0	défait(s)
T° aimant :	1.740	K
Pression satellite :	1.150	bar
Niveau satellite :	59.63	%
Pression Dewar :	1.250	bar
Pression Air/Azote :	82.030	bar
Vide Aimant :	7.6599998474	mbar
Synthèse électrique		
Détails process	0	défait(s)
Courant aimant :	1485.300	A
Présence 48V :	52.410	V
Statuts		
Compteur OB1	-19233	
Etat du vide :	OK	
Etat PCL :	ARRET	
Etat PCS :	MARCHE	
Présence 400V secours :	MARCHE	
Présence 400V normal :	MARCHE	



- <http://132.166.10.46:8080/dbwr/view.jsp?display=http://132.166.10.46/phoebus/ISEULT/AcquittementAlarmeFLS.bob>

Sum up & GUI technologies



<https://github.com/JeffersonLab/epics2web/wiki/Similar-Projects>

→ Reassuring that we are all (almost) going in a similar direction

GUI Situation & Strategy @ CEA IRFU

Technical criteria :

- Portability and interoperability (various OS of client workstations)
- User autonomy (Drag and Drop technology, with a GUI no command line)
- Modular and MVC design pattern (switch to any datasource separately from GUI)
- Reusable views and widgets
- A external support and maintenance of the solution
- Web client without software installation (not Java Web Start)

GUI guideline : Document defined at SOLEIL and at IRFU ([see guideline for CEA](#))

- Common layer definition
- Code color fixed in color.def configuration file for CS-STUDIO
- LSAP (Less Smart As Possible) , all the business and logical should be managed by server. (Avoid script in the client)
- Name the widget according to its function . Ex : Textfield_VoltageValue for a Textfield which displays a voltage value
- Reuse embedded display and defined MACRO or parameters

Future strategy and development (POC) :

Loïc > Web solutions analysis (data source separation, Phoebe format reusability ...)

- Taranta (MaxIV), MXCube ESRF, Diamond (cs-web-proto) , BNL (DBWR)

Katy > CS-Studio and tools around, roadmap , improvements ...

- Security, Muscade bridge , DBWR correction, replay mode, alarm service

Mathis, Katy , Loïc and CDD , apprentice, internship ...=> Executive developers



irfu

Any questions ?

Thank you for your attention